

Single-Agent vs. Multi-Agent LLM Systems: Evaluating Agentic Architectures for Reasoning Performance and Computational Efficiency

1. Introduction to the Research Topic

Large Language Models (LLMs) are increasingly being used in agentic systems where multiple AI agents collaborate to solve a problem. In a traditional single-agent system, one LLM receives a task, reasons about it, and produces a final response. In a multi-agent system, several LLM-based agents may independently solve the same problem, review each other's work, debate possible answers, or divide the task into smaller parts. These architectures are often expected to improve reasoning accuracy and reliability, but they also require more model calls, tokens, execution time, and computational cost.

The main goal of this project is to investigate whether multi-agent LLM systems actually provide enough reasoning improvement to justify their additional complexity and resource usage. One important issue is that multi-agent systems may appear better simply because they use more total computation than a single agent. Therefore, this project will compare different agent architectures both under normal conditions and under similar computational budgets.

The primary research question is:

Under comparable computational budgets, when do multi-agent LLM systems outperform single-agent systems, and when does adding more agents lead to diminishing or negative returns?

The project will also examine several related questions:

- Does increasing the number of agents consistently improve reasoning accuracy?
- Are some multi-agent architectures more effective than others?
- Do difficult reasoning tasks benefit more from multiple agents than simpler tasks?
- How much additional cost, latency, and token usage is required for each improvement in performance?
- Can collaboration between agents sometimes reduce the quality of a correct answer?

The expected outcome is a better understanding of when multi-agent reasoning is useful and when a simpler single-agent approach may be more efficient. Rather than assuming that more agents automatically produce better results, this project will experimentally evaluate the trade-off between reasoning performance and computational cost.

2. Detailed Plan to Complete the Research

The project will be implemented primarily in **Python using Google Agent Development Kit (Google ADK)**. Google ADK will be used as the framework for building and controlling different agentic structures while keeping the underlying language model, prompts, datasets, and evaluation process as consistent as possible. This will make it easier to determine whether differences in performance are caused by the agent architecture itself rather than unrelated implementation differences.

Several agent configurations will be developed. The first will be a **single-agent baseline**, where one LLM independently solves each task. This baseline will be compared with multiple multi-agent structures, including:

- **Parallel voting:** several agents independently solve the same problem and the final answer is selected through majority voting or an evaluator.
- **Sequential review:** one agent produces an answer and another agent reviews and improves it.
- **Multi-agent debate:** agents generate different solutions, critique each other, and then produce a final answer.
- **Coordinator-worker structure:** if time permits, one agent will divide a task into smaller parts and delegate them to other agents.

The experiments will also vary the number of agents, such as one, two, three, and five agents. This will allow the project to study whether performance continues to improve as more agents are added or whether the benefits eventually become too small compared with the additional resources required.

The systems will be evaluated using established reasoning benchmarks rather than manually created questions. Possible datasets include **GSM8K** for mathematical reasoning, **MMLU-Pro** for advanced knowledge and reasoning, and selected tasks from **BIG-Bench Hard (BBH)** for logical and multi-step reasoning. A manageable subset of these benchmarks will be used so that the experiments remain feasible within the semester. The same questions will be given to every architecture to ensure a fair comparison.

Two main types of experiments will be conducted. First, the architectures will be compared under normal conditions, where multi-agent systems are allowed to use the additional tokens and model calls they naturally require. Second, a controlled experiment will compare systems under approximately equal computational or token budgets. This second experiment is especially important because it will help determine whether any improvement comes from actual collaboration between agents or simply from using more total inference-time computation.

The systems will be evaluated using several quantitative metrics:

- Reasoning accuracy

- Total token usage
- Number of model calls
- Execution latency
- Estimated API or computational cost
- Accuracy improvement per additional agent
- Accuracy improvement relative to additional token usage

The project will also analyze failure cases. For example, it will examine situations where a single agent initially produces the correct answer but additional agents cause the final result to become incorrect. These cases may reveal problems such as agents reinforcing incorrect reasoning, following an incorrect majority, or unnecessarily changing a correct solution.

After the experiments are completed, the results will be presented using tables and graphs comparing accuracy, cost, token usage, latency, and number of agents. The analysis will focus on identifying patterns such as diminishing returns and determining which architectures provide the best balance between reasoning quality and computational efficiency.

The final deliverables will include the Google ADK-based experimental framework, the implemented single-agent and multi-agent architectures, benchmark evaluation scripts, collected experimental results, comparison graphs, and a final research report. The report will summarize the methodology, findings, limitations, and conclusions about when multi-agent LLM systems provide meaningful benefits over single-agent approaches.

Overall, the project goes beyond simply building an AI agent system. **Google ADK will be used as the experimental platform, while the main research contribution will be the systematic evaluation of how different agentic architectures affect reasoning performance, cost, and computational efficiency.**